

# Requirement Analysis

## CampusCare – Smart College Complaint Management System

*Understanding what needs to be built before deciding how to build it.*

| Document Information | Details                                                                                        |                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Project              | CampusCare – Smart College Complaint Management System                                         |                                                                                                                      |
| Document             | Requirement Analysis                                                                           |                                                                                                                      |
| Primary scope        | College complaint submission, tracking, assignment, resolution and monitoring                  |                                                                                                                      |
| Prepared for         | Advanced Software Engineering & AI foundation                                                  |                                                                                                                      |
| Program Code         | CE3                                                                                            |                                                                                                                      |
| Batch Code           | 8B                                                                                             |                                                                                                                      |
| Group no.            | Group 2                                                                                        |                                                                                                                      |
| Team size            | 7 members                                                                                      |                                                                                                                      |
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# 1. Position of Requirement Analysis in the Project Lifecycle

Requirement analysis is the first major activity in the project planning and design phase. For CampusCare, it establishes a shared understanding of the college complaint-management problem before system design, database design, sprint planning and development begin.

| Stage                    | CampusCare Example                                                                                  |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| 1. Business Problem      | Campus complaints are difficult to report, track, assign and monitor through fragmented channels.   |
| 2. Requirement Analysis  | Define users, complaint workflow, business rules, functional requirements and quality expectations. |
| 3. System Design         | Define application components, interfaces, workflows and responsibilities.                          |
| 4. Database Design       | Define complaint, user, department, category, history, notification and feedback data.              |
| 5. Agile Sprint Planning | Break approved requirements into implementable user stories and tasks.                              |

## 2. Case Study – Smart College Complaint Management System

### 2.1 Business Problem

Students and staff may encounter problems involving classrooms, laboratories, Wi-Fi, hostels, maintenance, library facilities, transportation, security, cleanliness and other campus services. When these issues are reported through informal or disconnected channels, complaints can be lost, duplicated, delayed or assigned to the wrong team.

### 2.2 Current Problems

- No single, centralized record of complaints.
- Students have limited visibility into complaint status and progress.
- Departments may receive complaints through multiple channels without a common tracking process.
- Urgent complaints may not be escalated consistently.
- Management has limited visibility into complaint volume, categories and resolution performance.
- Manual classification and assignment can consume staff time.
- There may be no complete history of status changes and actions taken.

### 2.3 Business Goal

Build a centralized complaint-management system that allows students to report campus issues, enables responsible departments to receive and resolve complaints, gives administrators visibility into operations, and provides automated notifications and escalation for timely resolution.

## 3. Stakeholders

A stakeholder is a person, group or organization that is interested in the system or affected by it.

| Stakeholder            | Interest / Need                                                                       |
|------------------------|---------------------------------------------------------------------------------------|
| Students               | Report campus problems, track their complaints and provide feedback after resolution. |
| Department Staff       | View assigned complaints, investigate issues, update status and record resolution.    |
| Department Managers    | Monitor department workload, assign/reassign complaints and handle escalations.       |
| College Administrators | Manage users and configurations, monitor complaints and view system-wide reports.     |

|                                 |                                                                       |
|---------------------------------|-----------------------------------------------------------------------|
| College Management              | Monitor service quality, complaint trends and resolution performance. |
| System/Automation Administrator | Maintain automated notification and escalation workflows.             |

## 4. Actors

An actor is a user or external system that directly interacts with the application.

### 4.1 Primary Actors

| Actor              | Description                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------|
| Student            | Creates complaints, tracks submitted complaints, comments, receives updates and submits feedback. |
| Department Staff   | Handles assigned complaints and updates progress and resolution information.                      |
| Department Manager | Supervises departmental complaints, assigns work and handles escalation.                          |
| Administrator      | Manages users, departments, categories and system-wide reporting.                                 |

### 4.2 Supporting / External Actors

| Actor/System           | Interaction                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------|
| AI Analysis Service    | Analyzes complaint text and returns a suggested category, priority and summary.                       |
| n8n Automation Service | Receives events or scheduled triggers and performs notifications, reminders and escalation workflows. |

## 5. Functional Requirements

A functional requirement describes what the system should do. The following requirements are written so that they can later be traced to user stories, design components, database entities, development tasks and tests.

| ID    | Requirement                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | The system shall allow a registered student to log in and access the student functions available to that role.                |
| FR-02 | The system shall allow a student to create a complaint with a title, description and relevant location/details.               |
| FR-03 | The system shall allow a student to attach supporting images/files subject to configured validation rules.                    |
| FR-04 | The system shall generate a unique complaint identifier for every successfully created complaint.                             |
| FR-05 | The system shall assign an initial status to a newly created complaint.                                                       |
| FR-06 | The system shall allow students to view the complaints they have submitted.                                                   |
| FR-07 | The system shall display the current status and history of a student's complaint.                                             |
| FR-08 | The system shall allow authorized staff to view complaints assigned to them or their permitted department.                    |
| FR-09 | The system shall allow authorized managers to assign or reassign complaints to department staff.                              |
| FR-10 | The system shall allow authorized staff to update complaint status according to the permitted workflow.                       |
| FR-11 | The system shall record status changes, the responsible user, timestamp and relevant remarks.                                 |
| FR-12 | The system shall allow staff to add comments and resolution details to a complaint.                                           |
| FR-13 | The system shall allow authorized staff to mark a complaint as resolved when the required resolution information is provided. |
| FR-14 | The system shall allow a student to close a resolved complaint or provide feedback according to the defined workflow.         |

|       |                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------|
| FR-15 | The system shall allow a student to reopen an eligible resolved complaint according to the configured business rule. |
| FR-16 | The system shall support complaint categories and map categories to responsible departments.                         |
| FR-17 | The system shall support priority levels for complaints.                                                             |
| FR-18 | The system shall use AI-assisted analysis to suggest complaint category, priority and a concise summary.             |
| FR-19 | The system shall store the AI-generated analysis associated with the complaint.                                      |
| FR-20 | The system shall allow authorized users to review or override AI suggestions when required.                          |
| FR-21 | The system shall notify relevant users when important complaint events occur.                                        |
| FR-22 | The system shall support automated reminders and escalation for overdue or critical complaints.                      |
| FR-23 | The system shall provide administrators with complaint statistics and operational reports.                           |
| FR-24 | The system shall provide filtering/searching by fields such as status, category, priority, department and date.      |
| FR-25 | The system shall allow students to submit a rating and optional comments for an eligible resolved complaint.         |
| FR-26 | The system shall restrict features and data according to the authenticated user's role and permissions.              |
| FR-27 | The system shall allow administrators to manage users, departments and complaint categories.                         |

## 6. Non-Functional Requirements

A non-functional requirement describes how the system should behave or the constraints under which it should operate.

| ID     | Quality Attribute       | Requirement                                                                                                                                                                               |
|--------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Response Time           | For normal application operations, the system should respond within an acceptable time for the project's expected user load; target approximately 2 seconds for common non-AI operations. |
| NFR-02 | Security                | The system shall authenticate users and enforce role-based authorization so users can access only permitted functions and data.                                                           |
| NFR-03 | Data Protection         | Passwords and sensitive authentication information shall not be stored in plain text; input shall be validated on the server.                                                             |
| NFR-04 | Availability            | The application should be available during normal college operating periods, subject to the hosting environment.                                                                          |
| NFR-05 | Reliability             | A successfully submitted complaint shall receive a unique identifier and shall not be silently lost.                                                                                      |
| NFR-06 | Auditability            | Important complaint status changes and assignment actions shall be traceable through history records.                                                                                     |
| NFR-07 | Scalability             | The design should allow additional departments, categories and users without requiring major changes to the complaint workflow.                                                           |
| NFR-08 | Usability               | Students and staff should be able to complete common tasks with clear forms, labels, validation messages and understandable status indicators.                                            |
| NFR-09 | Maintainability         | Business logic should be separated into appropriate application/service components so changes can be made without modifying unrelated features.                                           |
| NFR-10 | Testability             | Core business rules and service behavior shall be structured so they can be verified using automated tests.                                                                               |
| NFR-11 | Integration Reliability | Failures in external AI or automation services should not prevent the application from retaining the original complaint.                                                                  |
| NFR-12 | File Validation         | Uploaded files shall be restricted by permitted type and size to reduce security and storage risks.                                                                                       |

## 7. Functional vs Non-Functional Requirements

| Functional – What?          | Non-Functional – How well / under what constraints?                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Create a complaint          | The complaint form should be clear and validate required fields.                                                      |
| Assign a complaint          | Only authorized roles should be able to assign/reassign complaints.                                                   |
| Change complaint status     | Status changes must be recorded with a timestamp and responsible user.                                                |
| Generate a report           | Common reports should return within an acceptable response time for the expected data volume.                         |
| Send a notification         | Notification failure should not cause the original complaint transaction to be lost.                                  |
| Use AI for classification   | AI should be treated as an assistive service and the application must remain usable if AI is temporarily unavailable. |
| Escalate overdue complaints | Escalation should execute reliably according to the configured schedule/SLA rules.                                    |

## 8. Quick Classification Examples

| Statement                                                                                     | Classification                          |
|-----------------------------------------------------------------------------------------------|-----------------------------------------|
| The student should be able to create a complaint.                                             | Functional                              |
| The application should respond within approximately 2 seconds for common operations.          | Non-Functional                          |
| The administrator should be able to deactivate a user.                                        | Functional                              |
| Only authorized administrators should be able to manage users.                                | Non-Functional / Security constraint    |
| The staff member should be able to update complaint status.                                   | Functional                              |
| The system should retain an audit trail of status changes.                                    | Functional with auditability constraint |
| The system should continue accepting complaints if the AI service is temporarily unavailable. | Non-Functional / Reliability            |
| The system should automatically escalate overdue complaints.                                  | Functional                              |

## 9. User Stories

In Agile development, requirements are often represented as user stories. The common format is:

**As a [user], I want [functionality], so that [benefit].**

| Actor   | User Story                                                                                                 |
|---------|------------------------------------------------------------------------------------------------------------|
| Student | As a student, I want to create a complaint so that I can report a campus problem and request assistance.   |
| Student | As a student, I want to view my complaint status so that I know whether my issue is being handled.         |
| Student | As a student, I want to see complaint history so that I can understand the actions taken on my issue.      |
| Student | As a student, I want to attach evidence to a complaint so that staff can better understand the problem.    |
| Student | As a student, I want to provide feedback after resolution so that the college can measure service quality. |

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Department Staff      | As a department staff member, I want to view assigned complaints so that I can work on the issues under my responsibility.           |
| Department Staff      | As a department staff member, I want to update complaint status so that students and managers can track progress.                    |
| Department Staff      | As a department staff member, I want to record resolution details so that the completed work is documented.                          |
| Department Manager    | As a department manager, I want to assign complaints to staff so that each issue has a responsible person.                           |
| Department Manager    | As a department manager, I want to see overdue complaints so that I can take corrective action.                                      |
| Administrator         | As an administrator, I want to manage users, departments and categories so that the complaint system reflects the college structure. |
| Administrator         | As an administrator, I want to view complaint reports so that I can identify trends and service bottlenecks.                         |
| Administrator         | As an administrator, I want to monitor escalated complaints so that critical or overdue issues receive attention.                    |
| Student               | As a student, I want the system to suggest a complaint category and priority so that my complaint can be routed more quickly.        |
| Administrator/Manager | As an administrator or manager, I want to review AI suggestions so that incorrect classifications can be corrected.                  |
| College Management    | As management, I want to see complaint trends and resolution performance so that I can improve campus services.                      |

## 10. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be considered complete and acceptable.

### 10.1 Create Complaint

**As a student, I want to create a complaint so that I can report a campus problem and request assistance.**

| # | Acceptance Criterion                                                      |
|---|---------------------------------------------------------------------------|
| 1 | The student must be authenticated.                                        |
| 2 | The student must provide a complaint title.                               |
| 3 | The student must provide a meaningful complaint description.              |
| 4 | The system must validate required fields before submission.               |
| 5 | The system must create the complaint when valid information is submitted. |
| 6 | The system must generate a unique complaint number.                       |
| 7 | The new complaint must receive the defined initial status.                |
| 8 | The complaint must be associated with the submitting student.             |
| 9 | The student must be able to view the newly created complaint.             |

### 10.2 View Assigned Complaint

**As a department staff member, I want to view assigned complaints so that I can work on the issues under my responsibility.**

| # | Acceptance Criterion                                                                          |
|---|-----------------------------------------------------------------------------------------------|
| 1 | The staff member must be authenticated.                                                       |
| 2 | The staff member must see only complaints they are authorized to access.                      |
| 3 | The complaint list must display complaint number, title, priority and status.                 |
| 4 | The staff member must be able to open a complaint detail page.                                |
| 5 | The detail page must show relevant description, category, assignment and history information. |

### 10.3 Assign Complaint

As a department manager, I want to assign complaints to staff so that each issue has a responsible person.

| # | Acceptance Criterion                                                                |
|---|-------------------------------------------------------------------------------------|
| 1 | The manager must be authenticated and authorized to assign complaints.              |
| 2 | The complaint must exist and be in a state that permits assignment.                 |
| 3 | The selected staff member must belong to the permitted department.                  |
| 4 | The assignment must be saved successfully.                                          |
| 5 | The complaint history must record the assignment action.                            |
| 6 | The relevant staff member must receive a notification or queued notification event. |

### 10.4 AI-Assisted Classification

As a student, I want the system to suggest a complaint category and priority so that my complaint can be routed more quickly.

| # | Acceptance Criterion                                                                                |
|---|-----------------------------------------------------------------------------------------------------|
| 1 | The original complaint text must be retained before AI analysis.                                    |
| 2 | The system should send the relevant complaint text to the AI analysis service when available.       |
| 3 | The AI response should contain a suggested category and priority in the expected structured format. |
| 4 | The system should store the AI-generated summary and suggestions.                                   |
| 5 | AI failure or timeout must not delete or invalidate the original complaint.                         |
| 6 | An authorized user must be able to review or override the suggestion.                               |

### 10.5 Escalated Complaints

As an administrator, I want to monitor escalated complaints so that critical or overdue issues receive attention.

| # | Acceptance Criterion                                                               |
|---|------------------------------------------------------------------------------------|
| 1 | The system must identify complaints that meet the configured escalation condition. |
| 2 | The escalation action must be recorded.                                            |
| 3 | The complaint status must reflect escalation where defined by the workflow.        |
| 4 | The relevant manager/administrator must receive a notification event.              |
| 5 | The escalated complaint must appear in the appropriate dashboard/filter.           |

## 11. Given – When – Then

Acceptance criteria can also be expressed using a simple Given–When–Then structure.

### 11.1 Create Complaint

| Given                                                             | When                                                                        | Then                                                                                                                   |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| The student is logged in and has permission to submit complaints. | The student enters a valid title and description and submits the complaint. | The system creates a complaint, generates a unique number, sets the initial status and associates it with the student. |

### 11.2 AI Classification

| Given | When | Then |
|-------|------|------|
|-------|------|------|

|                                                                     |                                                   |                                                                                                          |
|---------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| A valid complaint has been created and the AI service is available. | The system sends the complaint text for analysis. | The system stores the returned category, priority and summary and makes the result available for review. |
|---------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|

### 11.3 Overdue Escalation

| Given                                                                                | When                                                         | Then                                                                                                                                |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| A complaint has exceeded its configured resolution threshold and remains unresolved. | The scheduled automation workflow checks overdue complaints. | The complaint is marked/escalated according to the business rule and a notification event is generated for the responsible manager. |

### 11.4 Unauthorized Access

| Given                                                                         | When                                                  | Then                                                         |
|-------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------|
| A user does not have permission to access another role's restricted function. | The user attempts to access the restricted operation. | The system denies access and does not expose protected data. |

## 12. From Requirement to Development

Requirement analysis creates the foundation for the work that follows. Each important requirement should be traceable through design, database work, development tasks and testing.

| Stage                | CampusCare Example                                                                                   |
|----------------------|------------------------------------------------------------------------------------------------------|
| Business Requirement | Student can report a campus problem through a centralized complaint system.                          |
| User Story           | As a student, I want to create a complaint so that I can report a campus problem.                    |
| Acceptance Criteria  | Title and description required; complaint created; unique number generated; initial status assigned. |
| System Design        | Complaint workflow, controller/API, application service and authorization responsibilities.          |
| Database Design      | Complaint entity/table and relationships to User, Category, Department and History.                  |
| Development Task     | Implement Create Complaint feature, validation, persistence and automated tests.                     |
| Sprint               | Schedule the feature with related complaint-management work in an appropriate sprint.                |

## What Comes Next?

The approved requirements in this document become the input for the next project-planning activities. The team should progressively move from what needs to be built to how it will be designed and delivered.

| Next Module           | Expected Output from This Requirement Analysis                                  |
|-----------------------|---------------------------------------------------------------------------------|
| System Design         | Architecture, components, interfaces, workflows and role boundaries.            |
| Database Design       | Entities, relationships, keys, constraints and data-flow decisions.             |
| Agile Sprint Planning | Prioritized user stories, tasks, estimates and sprint assignments.              |
| Development           | Implementation of approved features and validation against acceptance criteria. |
| Testing               | Verification that functional and non-functional requirements are satisfied.     |